

Cumulative Frequency

Cumulative frequency is a running total. Plot at the END of each class, join with a smooth curve, then read across and down.



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THE TOPIC ON ONE MAP

Cumulative Frequency

01 Tables and graphs

running total

plot at upper bound

smooth curve

02 Median, quartiles, IQR

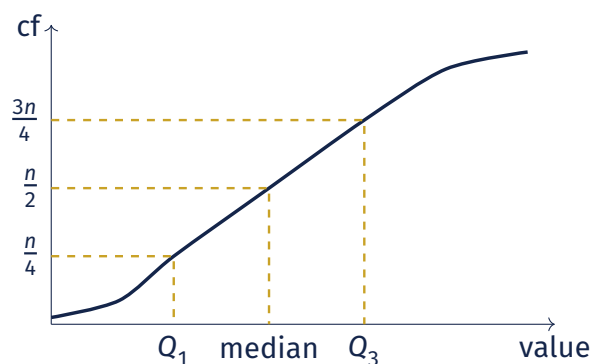
read at $\frac{n}{2}$ $\frac{n}{4}, \frac{3n}{4}$

03 More than / less than

read up, subtract

percentage, probability

READ THE GRAPH Across from the cumulative frequency, down to the value:



$$\text{IQR} = Q_3 - Q_1$$

more than x : $n - \text{reading at } x$

01 Cumulative Frequency Tables and Graphs

Keep adding: each cumulative frequency is everything up to the end of that class. Plot each total at the upper class boundary, start at the lowest boundary with 0, and join with a smooth S-shaped curve.

WHEN YOU SEE → YOU DO

“complete the cumulative frequency table”

→ running total; the last one = the total

“draw a cumulative frequency graph”

→ plot (upper bound, cf)

the first class starts at 10, not 0

→ also plot (10, 0)

a cumulative table is given

→ plot it directly — no adding

Running total

Plot at upper bounds

Smooth curve

Read the graph

WORKED MODEL • JANUARY 2022 P1H

Frequencies 7, 26, 24, 14, 7, 2 for $0 < t \leq 10$, ..., $50 < t \leq 60$.

1 7, 33, 57, 71, 78, 80

running total

2 plot (10, 7), (20, 33), (30, 57), ..., (60, 80)

upper bounds

ANSWER median: read across at 40 $\Rightarrow \approx 23$ min

× **TRAP** Plotting at the midpoints. Cumulative frequency is always plotted at the END of the class.

IDEA 01

Practise: Cumulative Frequency Tables and Graphs

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Running total

CH 6 • TOPIC 3 2 marks

Frequencies 6, 14, 20, 7, 3 for $0 < t \leq 10$, $10 < t \leq 20$, ..., $40 < t \leq 50$. Write the cumulative frequencies and the points you plot.

SELECTED PRACTICE

Q2

PRACTICE

Practice • Easy

CH 6 • TOPIC 3

5 marks

1 (a) The table shows information about the speeds of 100 lorries.

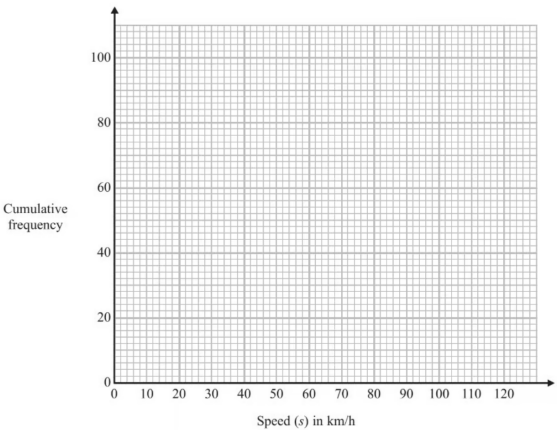
Speed (<i>s</i>) in km/h	Frequency
$0 < s \leq 20$	2
$20 < s \leq 40$	9
$40 < s \leq 60$	23
$60 < s \leq 80$	31
$80 < s \leq 100$	27
$100 < s \leq 120$	8

Complete the cumulative frequency table for this information.

Speed (<i>s</i>) in km/h	Cumulative frequency
$0 < s \leq 20$	2
$0 < s \leq 40$	
$0 < s \leq 60$	
$0 < s \leq 80$	
$0 < s \leq 100$	
$0 < s \leq 120$	

(1 mark)

(b) On the grid, draw a cumulative frequency graph for your table.



(2 marks)

(c) Find an estimate for the number of lorries with a speed of more than 90 km/h.

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 6 • TOPIC 3**2 marks**

- 9 20 students each record the mass, p grams, of their pencil case. The table below shows the results.

Mass (p grams)	$0 < p \leq 50$	$50 < p \leq 100$	$100 < p \leq 125$	$125 < p \leq 150$	$150 < p \leq 200$
Frequency	2	5	4	6	3

Use the frequency table to complete the cumulative frequency table.

Mass (p grams)	$p \leq 50$	$p \leq 100$	$p \leq 125$	$p \leq 150$	$p \leq 200$
Cumulative frequency					20

(2 marks)

Q4

PRACTICE

Practice • Medium

CH 6 • TOPIC 36 marks

4 (a) The table shows information about the times taken by 100 people in a fun run.

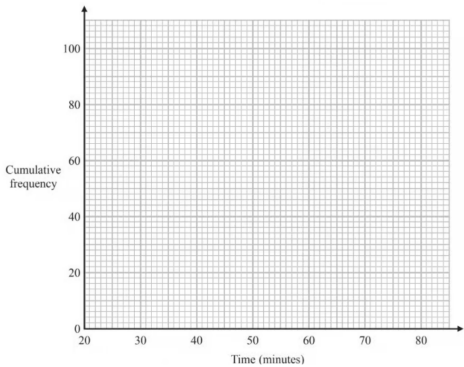
Time (t minutes)	Frequency
$20 < t \leq 30$	4
$30 < t \leq 40$	16
$40 < t \leq 50$	36
$50 < t \leq 60$	24
$60 < t \leq 70$	14
$70 < t \leq 80$	6

Complete the cumulative frequency table for this information.

Time (t minutes)	Cumulative frequency
$20 < t \leq 30$	
$20 < t \leq 40$	
$20 < t \leq 50$	
$20 < t \leq 60$	
$20 < t \leq 70$	
$20 < t \leq 80$	

(1 mark)

(b) On the grid, draw a cumulative frequency graph for your table.



(2 marks)

(c) Use your graph to find an estimate for the median time.

(1 mark)

(d) Use your graph to find an estimate for the number of people who took longer than 63 minutes.

(2 marks)

Q5

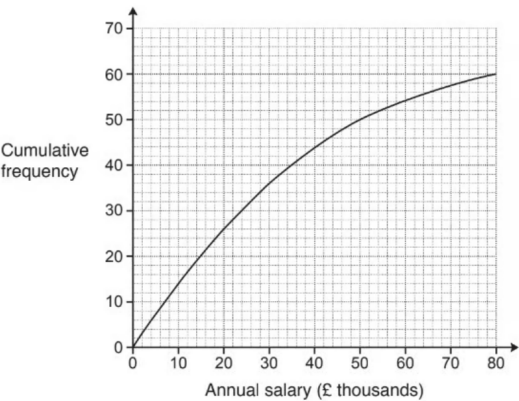
PRACTICE

Practice • Hard

CH 6 • TOPIC 39 marks

PRACTISE • 6.3

13 (a) The cumulative frequency graph summarises the annual salary, p (£ thousands), of the 60 workers in a factory.



Use the graph to estimate the median annual salary.

£ thousands
(1 mark)

(b) Complete this cumulative frequency table.

Annual salary, p (£ thousands)	Cumulative frequency
$p \leq 10$	
$p \leq 20$	
$p \leq 30$	
$p \leq 50$	
$p \leq 80$	

(2 marks)

(c) Use the information in the cumulative frequency table to calculate an estimate of the mean annual salary.

£ thousands
(5 marks)

(d) Explain why your estimate of the median is more reliable than your estimate of the mean.

(1 mark)

PAST PAPERS

Q6

PAST PAPER

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q11

CH 6 • TOPIC 3

5 marks

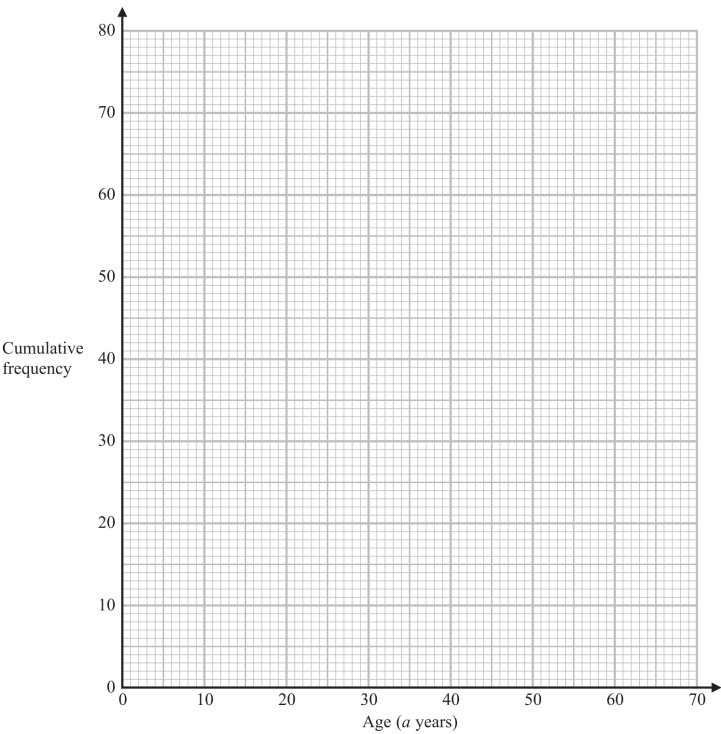
11 The frequency table gives information about the ages of the 80 people in a train carriage.

Age (a years)	Frequency
$0 < a \leq 20$	9
$20 < a \leq 30$	19
$30 < a \leq 40$	17
$40 < a \leq 50$	18
$50 < a \leq 60$	13
$60 < a \leq 70$	4

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(b) On the grid, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median age of the people in the train carriage.

..... years

(Total for Question 11 is 5 marks)

Q7

PAST PAPER

Edexcel 4MA1 • January 2021 • Paper 1HR • Q11

CH 6 • TOPIC 3 6 marks

- 11 The manager of a call centre asked the 120 people, who rang the call centre last week, how long they each waited before their call was answered.

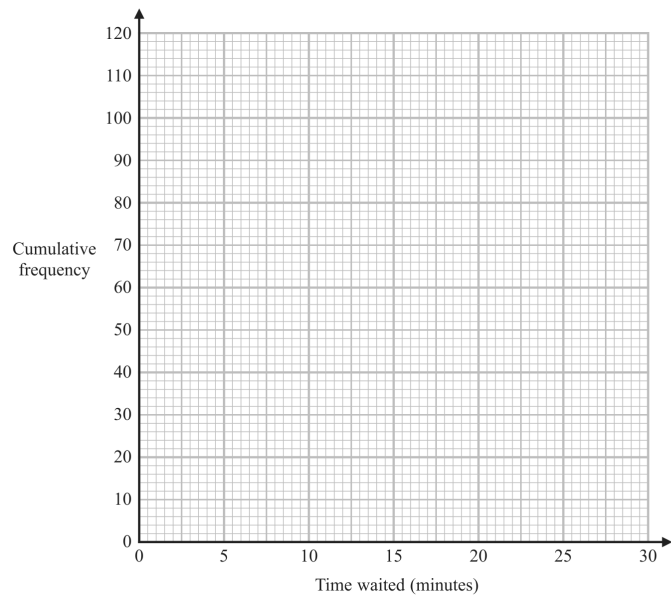
The table gives information about their replies.

Time waited (t minutes)	Frequency
$0 < t \leq 5$	8
$5 < t \leq 10$	15
$10 < t \leq 15$	17
$15 < t \leq 20$	28
$20 < t \leq 25$	33
$25 < t \leq 30$	19

- (a) Complete the cumulative frequency table.

Time waited (t minutes)	Cumulative frequency
$0 < t \leq 5$	
$0 < t \leq 10$	
$0 < t \leq 15$	
$0 < t \leq 20$	
$0 < t \leq 25$	
$0 < t \leq 30$	

- (b) On the grid below, draw a cumulative frequency graph for your table.



- (c) Use your graph to find an estimate for the median of the times waited.
..... minutes
- (d) Using your graph, find an estimate for the percentage of the 120 people who said that they waited longer than 23 minutes before their call was answered.
Show your working clearly.
.....%

(Total for Question 11 is 6 marks)

Q8

PAST PAPER

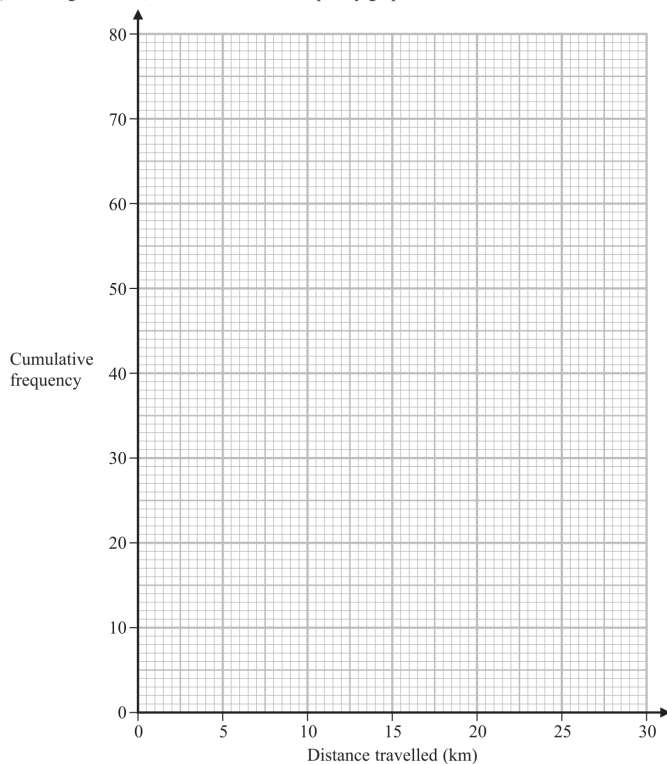
Edexcel 4MA1 • May 2021 • Paper 2H • Q12

CH 6 • TOPIC 3 7 marks

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 1.....
.....
.....
2.....
.....
.....

(Total for Question 12 is 7 marks)

Q9

PAST PAPER

Edexcel 4MA1 • November 2021 • Paper 1H • Q11

CH 6 • TOPIC 35 marks

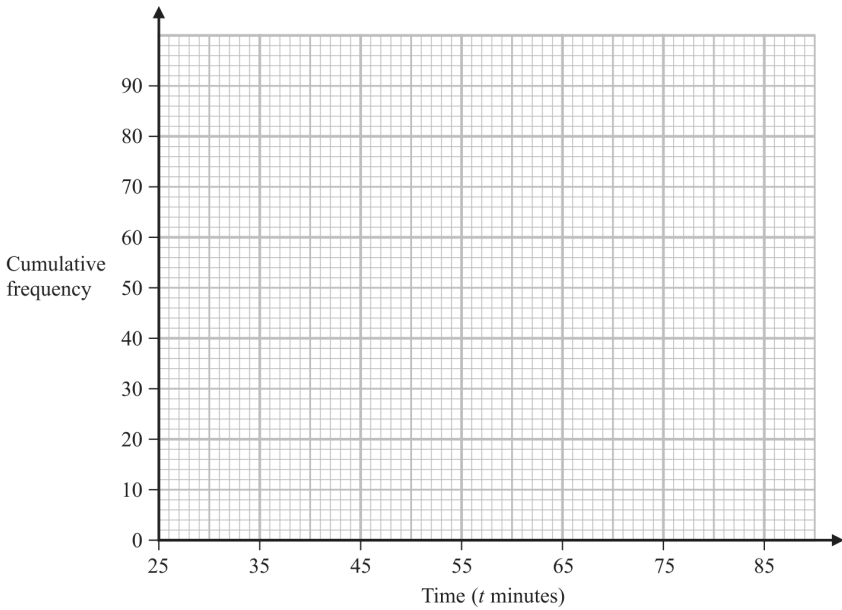
11 The table gives information about the times taken by 90 runners to complete a 10 km race.

Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(b) On the grid below, draw a cumulative frequency graph for your table.



Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

..... runners

(Total for Question 11 is 5 marks)

Q10

PAST PAPER

Edexcel 4MA1 • January 2022 • Paper 1H • Q12

CH 6 • TOPIC 36 marks

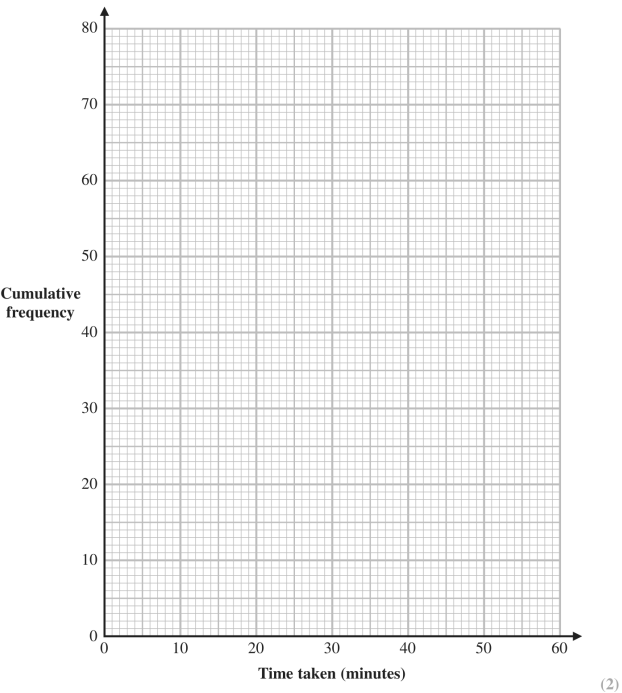
12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

Time taken (t minutes)	Frequency
$0 < t \leq 10$	7
$10 < t \leq 20$	26
$20 < t \leq 30$	24
$30 < t \leq 40$	14
$40 < t \leq 50$	7
$50 < t \leq 60$	2

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median time taken.

..... minutes

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

(Total for Question 12 is 6 marks)

Q11

PAST PAPER

Edexcel 4MA1 • June 2022 • Paper 2HR • Q11

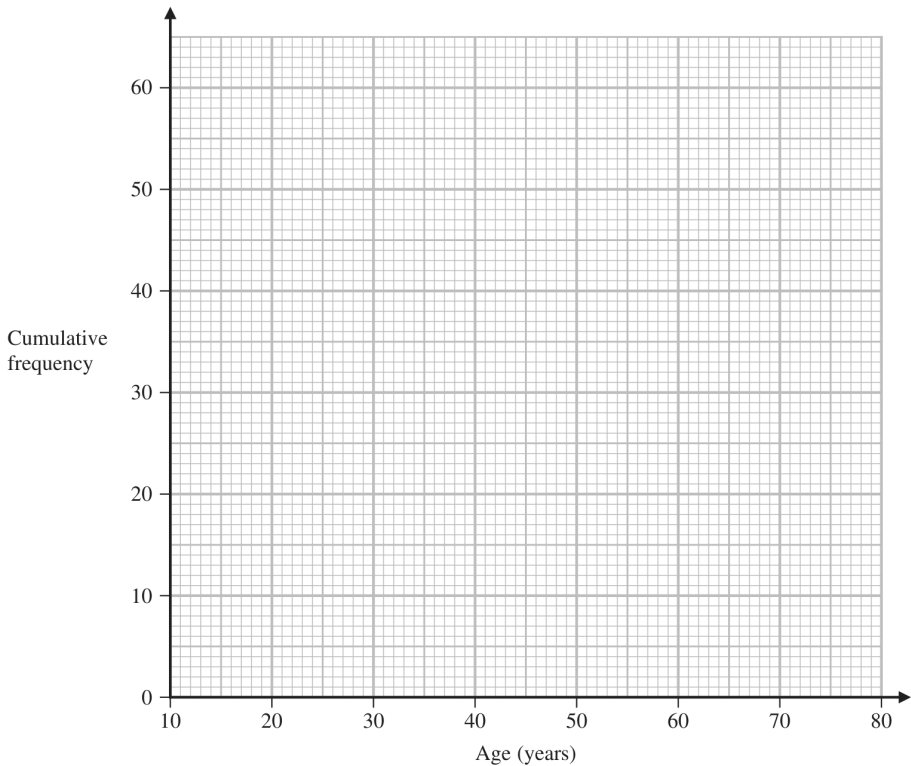
CH 6 • TOPIC 37 marks

PRACTISE • 6.3

11 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leq 20$	13
$10 < a \leq 30$	36
$10 < a \leq 40$	42
$10 < a \leq 50$	47
$10 < a \leq 60$	52
$10 < a \leq 70$	56
$10 < a \leq 80$	60

(a) On the grid, draw a cumulative frequency graph for the information in the table.



Question 11 continued

- (b) Use your graph to find an estimate for the median of the ages of these people.
..... years
- (c) Use your graph to find an estimate for the interquartile range of the ages of these people.
..... years
- (d) Use your graph to find an estimate for the number of these people who are older than 55 years.
.....

(Total for Question 11 is 7 marks)

Q12

PAST PAPER

Edexcel 4MA1 • June 2022 • Paper 2H • Q13

CH 6 • TOPIC 37 marks

13 The table gives information about the ages, in years, of 80 people in a train carriage.

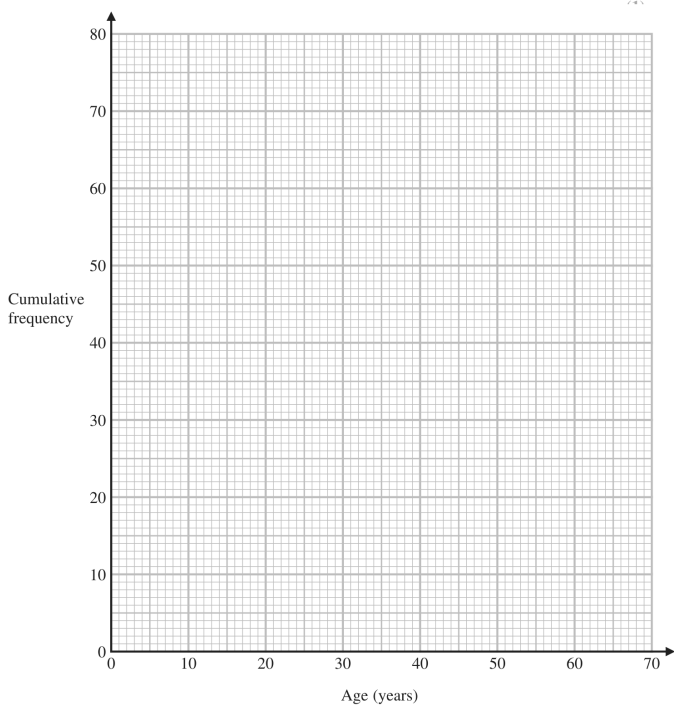
Age (a years)	Frequency
$0 < a \leq 20$	7
$20 < a \leq 30$	25
$30 < a \leq 40$	20
$40 < a \leq 50$	14
$50 < a \leq 60$	8
$60 < a \leq 70$	6

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(c) Use your graph to find an estimate for the median age of the 80 people.
..... years



Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

(d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.
.....

(Total for Question 13 is 7 marks)

Q13 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1HR • Q14

CH 6 • TOPIC 3 6 marks

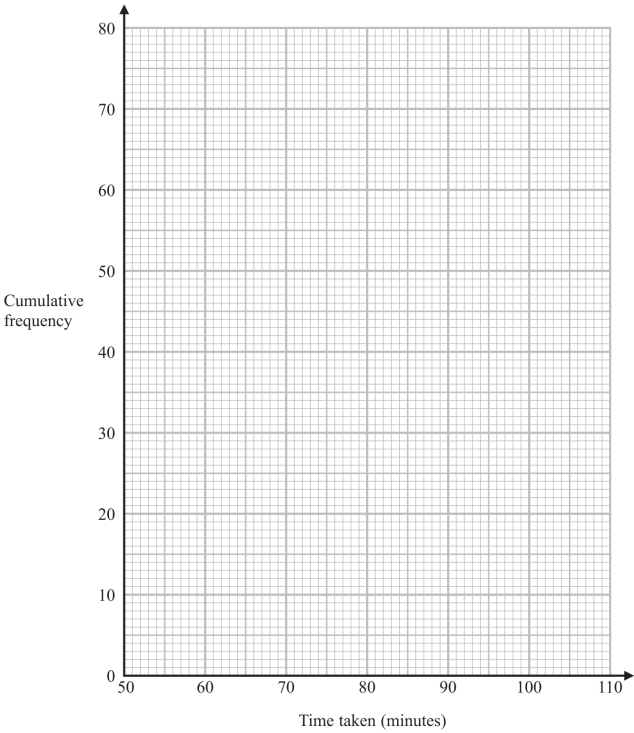
14 The table gives information about the times taken by 80 people to run a race.

Time taken (t minutes)	Frequency
$50 < t \leq 60$	15
$60 < t \leq 70$	16
$70 < t \leq 80$	21
$80 < t \leq 90$	14
$90 < t \leq 100$	8
$100 < t \leq 110$	6

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$50 < t \leq 60$	
$50 < t \leq 70$	
$50 < t \leq 80$	
$50 < t \leq 90$	
$50 < t \leq 100$	
$50 < t \leq 110$	

(b) On the grid below, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median time taken.

..... minutes

(d) Use your graph to find an estimate for the interquartile range of the times taken.

..... minutes

(Total for Question 14 is 6 marks)

Q14

PAST PAPER

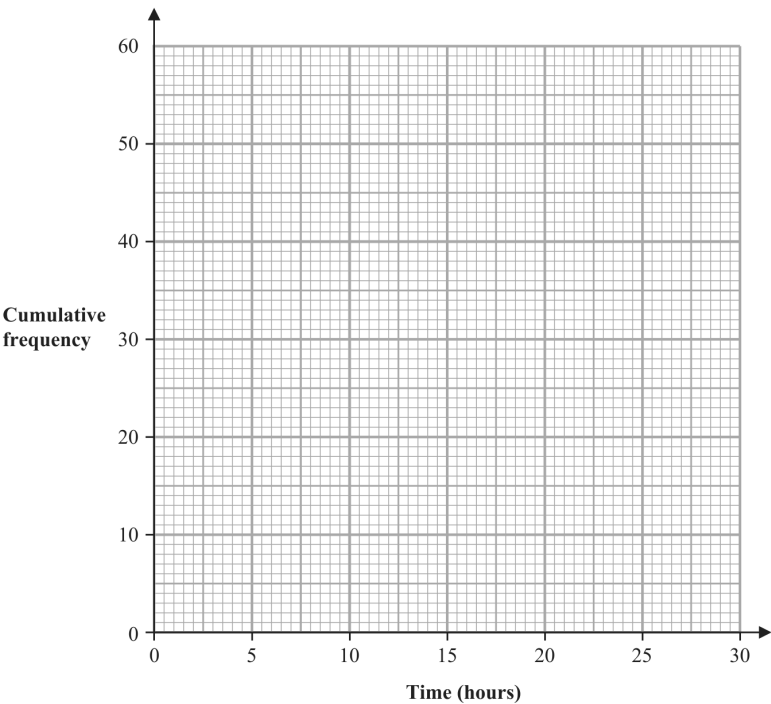
Edexcel 4MA1 • June 2024 • Paper 2HR • Q11

CH 6 • TOPIC 37 marks

11 The cumulative frequency table gives information about the time, in hours, that each of 60 workers spent working from home in one week.

Time (t hours)	Cumulative frequency
$0 < t \leqslant 5$	6
$0 < t \leqslant 10$	17
$0 < t \leqslant 15$	27
$0 < t \leqslant 20$	42
$0 < t \leqslant 25$	53
$0 < t \leqslant 30$	60

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(b) Use your graph to find an estimate for the interquartile range of the times.
..... hours

25 workers spent more than W hours working from home.

(c) Use your graph to find an estimate for the value of W
 $W =$

One of the 60 workers is chosen at random.
This worker spent H hours working from home.

(d) Find the probability that $5 < H \leqslant 10$
.....

(Total for Question 11 is 7 marks)

Q15

PAST PAPER

Edexcel 4MA1 • June 2024 • Paper 2H • Q13

CH 6 • TOPIC 37 marks

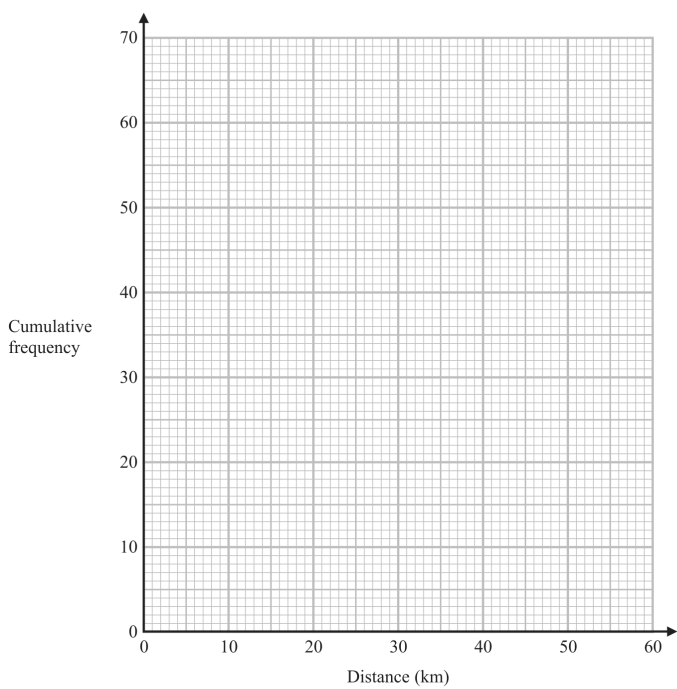
13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leq 10$	7
$10 < d \leq 20$	17
$20 < d \leq 30$	18
$30 < d \leq 40$	14
$40 < d \leq 50$	10
$50 < d \leq 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leq 10$	
$0 < d \leq 20$	
$0 < d \leq 30$	
$0 < d \leq 40$	
$0 < d \leq 50$	
$0 < d \leq 60$	

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the interquartile range of the distances.
..... km

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.
.....

(Total for Question 13 is 7 marks)

Q16

PAST PAPER

Edexcel 4MA1 • November 2024 • Paper 2H • Q13

CH 6 • TOPIC 3

6 marks

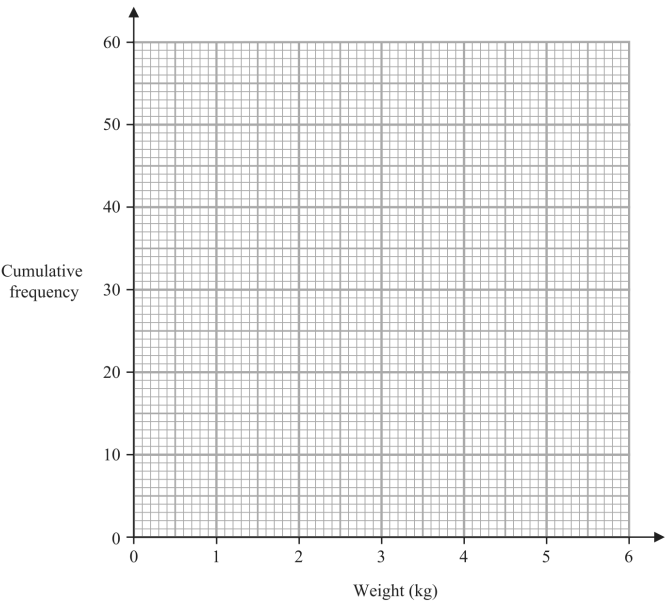
13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

(a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median weight of the 60 parcels.
..... kilograms

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.
.....

(Total for Question 13 is 6 marks)

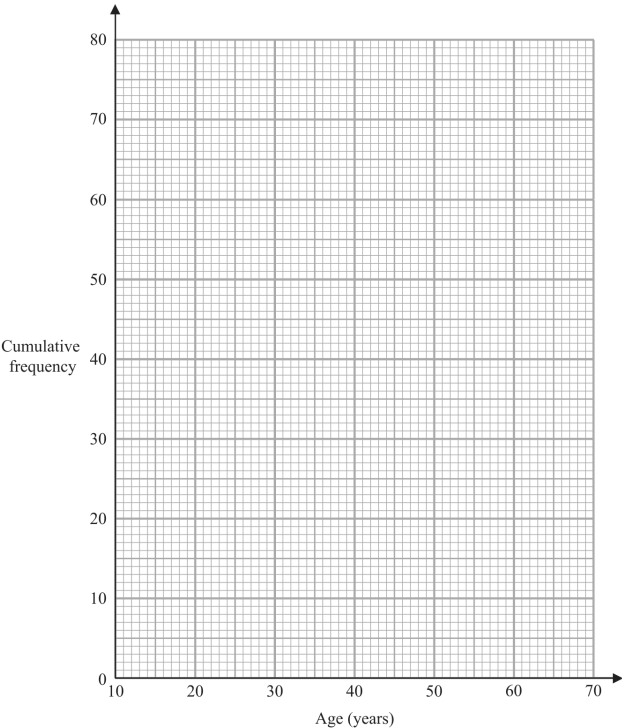
12 The table gives information about the ages of 80 people in a cinema.

Age (n years)	Frequency
$10 < n \leq 20$	12
$20 < n \leq 30$	15
$30 < n \leq 40$	20
$40 < n \leq 50$	18
$50 < n \leq 60$	9
$60 < n \leq 70$	6

(a) Complete the cumulative frequency table.

Age (n years)	Cumulative frequency
$10 < n \leq 20$	
$10 < n \leq 30$	
$10 < n \leq 40$	
$10 < n \leq 50$	
$10 < n \leq 60$	
$10 < n \leq 70$	

(b) On the grid below, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the percentage of the 80 people who are more than 46 years of age.
Give your answer correct to the nearest whole number.

.....%

(Total for Question 12 is 6 marks)

Q18

PAST PAPER

Edexcel 4MA1 • June 2025 • Paper 2H • Q13

CH 6 • TOPIC 36 marks

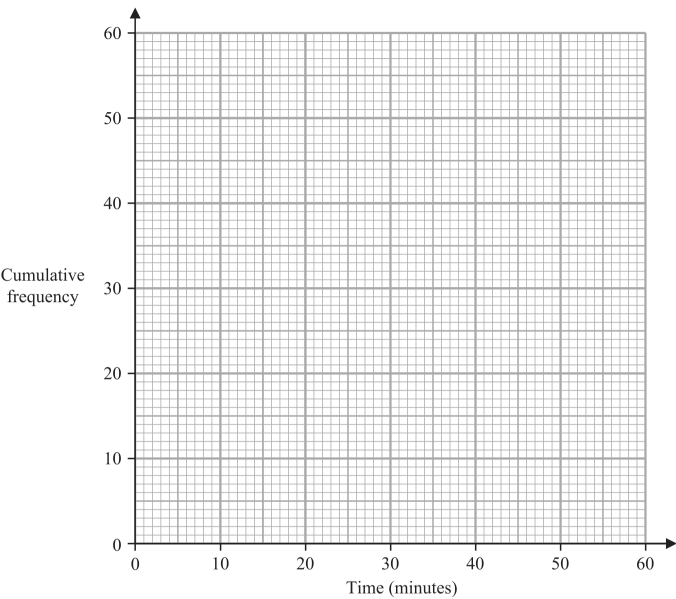
13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (<i>t</i> minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

(a) Complete the cumulative frequency table.

Time (<i>t</i> minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(c) Use your graph to find an estimate for the median time.

..... minutes

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

.....

(Total for Question 13 is 6 marks)

02 Median, Quartiles and IQR from the Graph

For n values on a graph: median at $\frac{n}{2}$, lower quartile at $\frac{n}{4}$, upper quartile at $\frac{3n}{4}$. Draw the lines across and down so the examiner sees them.

WHEN YOU SEE → YOU DO

“estimate the median”

→ across at $\frac{n}{2}$

“estimate the interquartile range”

→ readings at $\frac{n}{4}$ and $\frac{3n}{4}$, subtract

“compare the two hospitals”

→ median (average) and IQR (spread), in context

“is the range 40?”

→ No — the exact smallest and largest values are unknown

WORKED MODEL • JANUARY 2020 P1HR

60 people at Hospital A. Median and IQR?

1 across at 30 → ≈ 22.5 min

median

2 across at 15 and 45 → ≈ 16 and ≈ 28

quartiles

ANSWER IQR ≈ 12 min

× **TRAP** Reading the quartiles at 25 and 75 when $n = 60$. Use $\frac{1}{4}$ and $\frac{3}{4}$ of the total.

IDEA 02

Practise: Median, Quartiles and IQR from the Graph

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM Positions

CH 6 • TOPIC 3 2 marks

A cumulative frequency graph shows 80 people. At which cumulative frequencies do you read the median, Q_1 and Q_3 ?

SELECTED PRACTICE

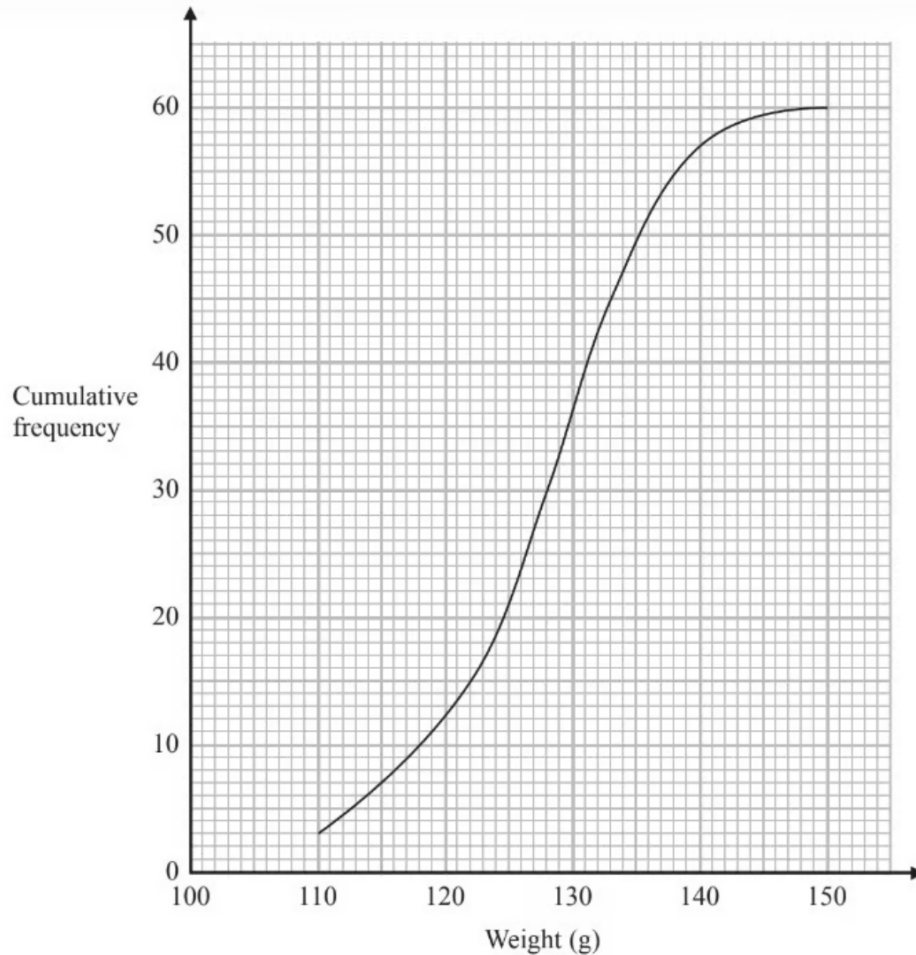
Q20**PRACTICE**

Practice • Easy

CH 6 • TOPIC 3

3 marks

6 (a) The cumulative frequency graph shows information about the weights of 60 apples.



Use the graph to find an estimate for the median weight.

(1 mark)

(b) Use the graph to find an estimate for the interquartile range of the weights.

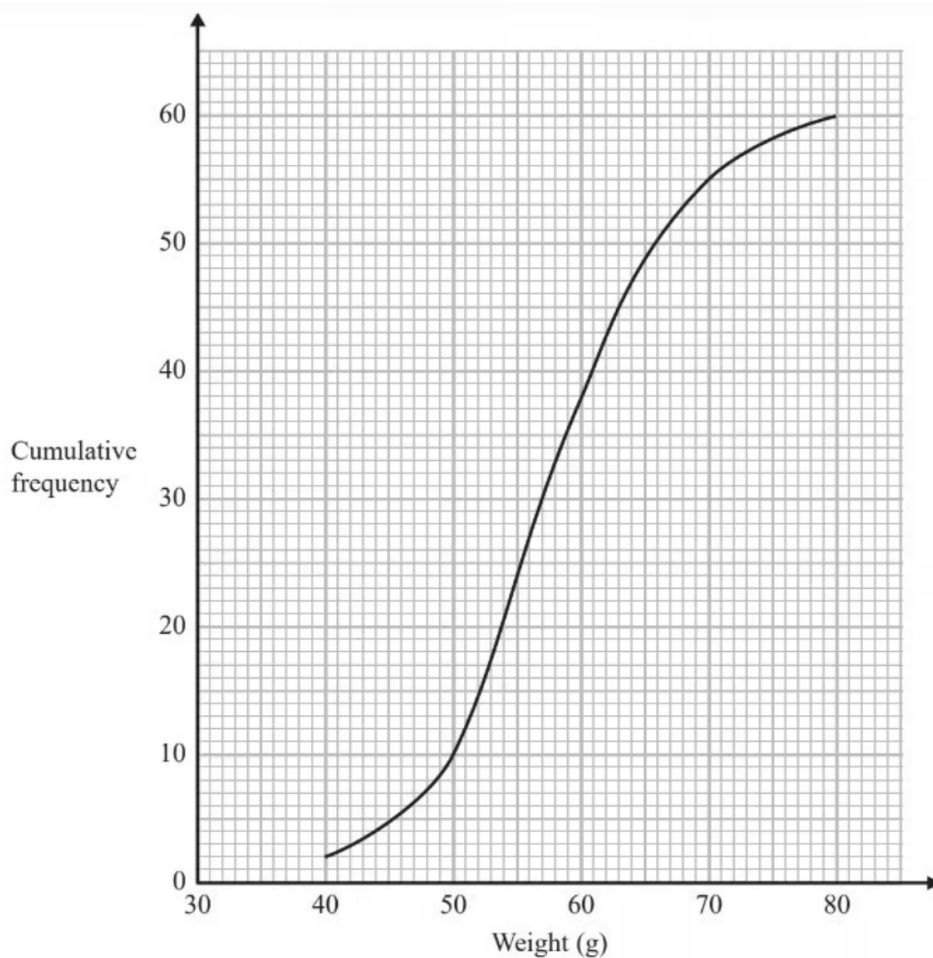
(2 marks)

Q21**PRACTICE**

Practice • Medium

CH 6 • TOPIC 3 4 marks

5 (a) The cumulative frequency graph shows information about the weights of 60 potatoes.



Use the graph to find an estimate for the median weight.

(1 mark)

- (b) Jamil says,
"80 - 40 = 40 so the range of the weights is 40 g."

Is Jamil correct?

You must give a reason for your answer.

(1 mark)

- (c) Show that less than 25% of the potatoes have a weight greater than 65 g.

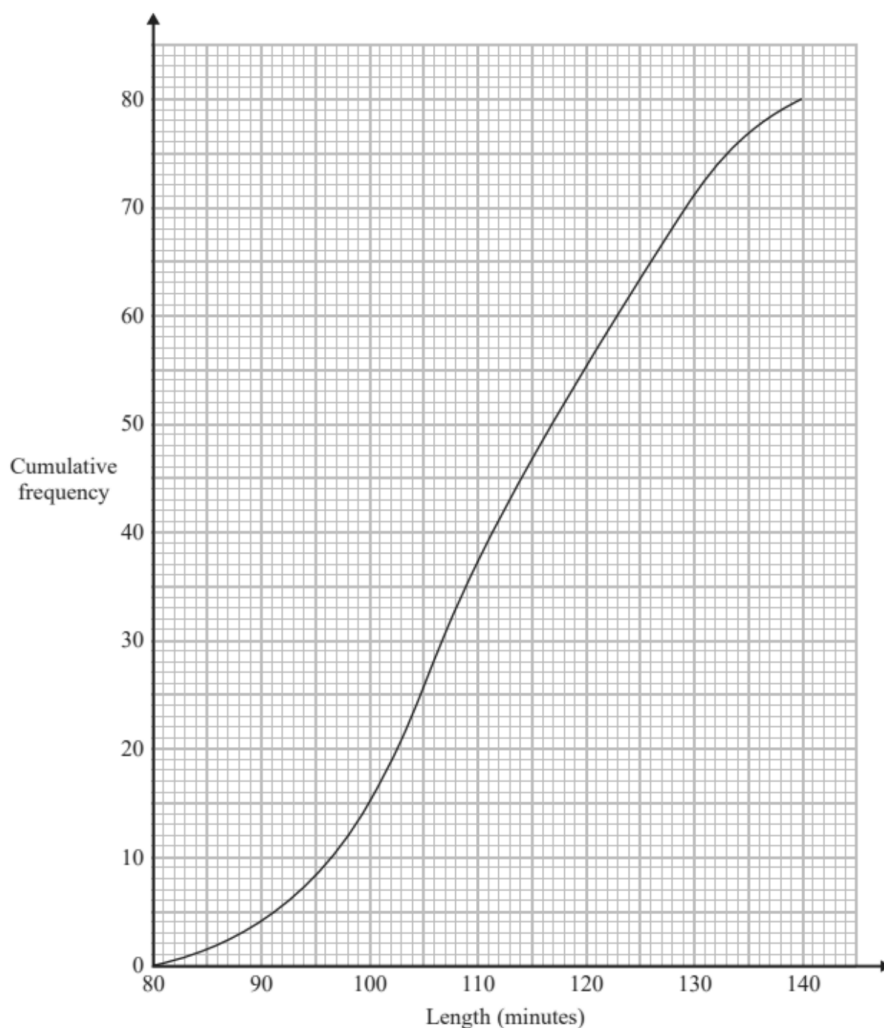
(2 marks)

Q22**PRACTICE**

Practice • Hard

CH 6 • TOPIC 3 5 marks

- 9 (a)** The cumulative frequency graph shows information about the length, in minutes, of each of 80 films.



Use the graph to find an estimate for the interquartile range.

..... minutes

(2 marks)

- (b)** Clare says,

"More than 35% of these films are over 120 minutes long."

Is Clare correct?

Give a reason for your answer.

(3 marks)

PAST PAPERS

Q23

PAST PAPER

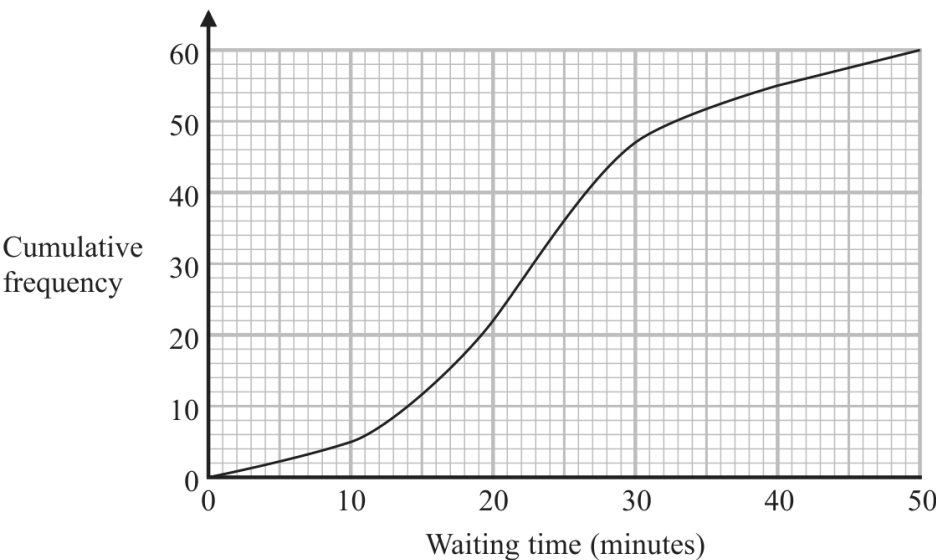
Edexcel 4MA1 • January 2020 • Paper 1HR • Q12

CH 6 • TOPIC 3

5 marks

PRACTISE - 6.3

12 The cumulative frequency graph gives information about the waiting times, in minutes, of people with appointments at Hospital A.



- (a) Use the graph to find an estimate of the median waiting time at Hospital A.
- minutes
- (b) Use the graph to find an estimate of the interquartile range of the waiting times at Hospital A.
- minutes

At a different hospital, Hospital B, the median waiting time is 28 minutes and the interquartile range of the waiting times is 19 minutes.

- (c) Compare the waiting times at Hospital A with the waiting times at Hospital B.

.....

.....

.....

.....

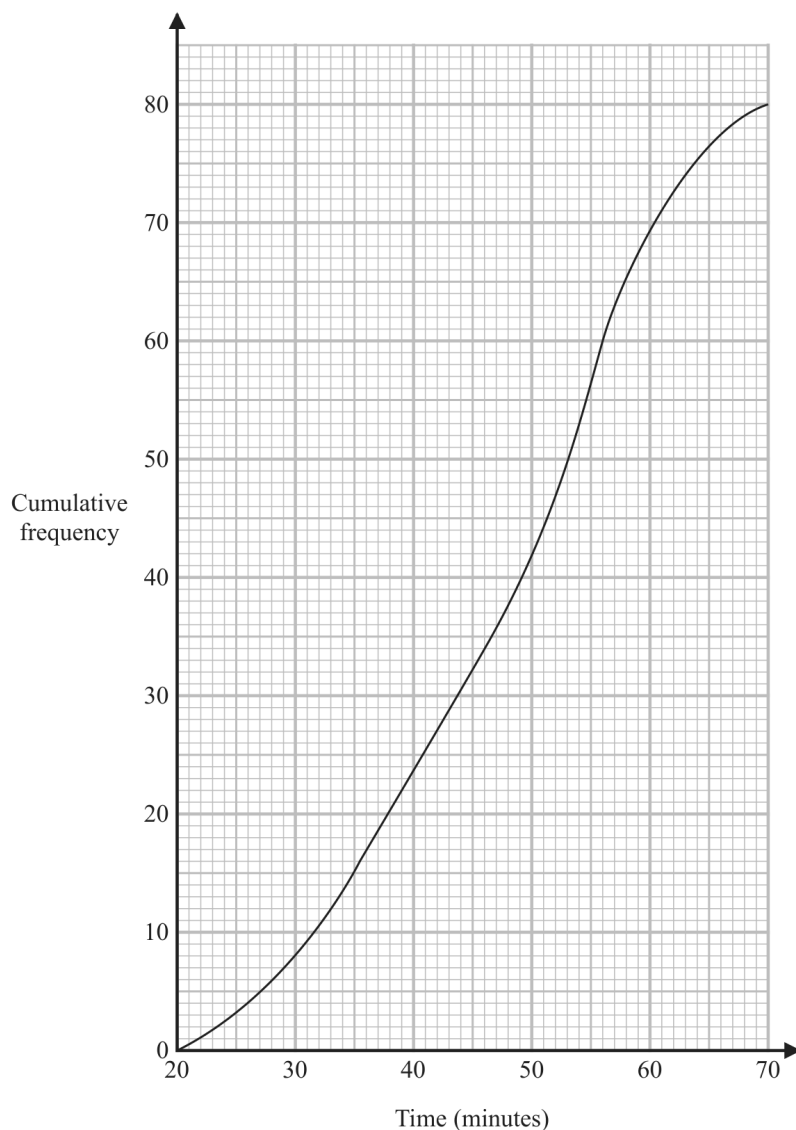
(Total for Question 12 is 5 marks)

Q24 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1H • Q12

CH 6 • TOPIC 3 5 marks

12 A total of 80 men and women took part in a race.

The cumulative frequency graph gives information about the times, in minutes, they took for the race.



(a) Use the graph to find an estimate for the interquartile range.

..... minutes

60% of the men took 50 minutes or less for the race.

No women took 50 minutes or less for the race.

(b) Work out an estimate for the number of men who took part in the race.

.....

(Total for Question 12 is 5 marks)

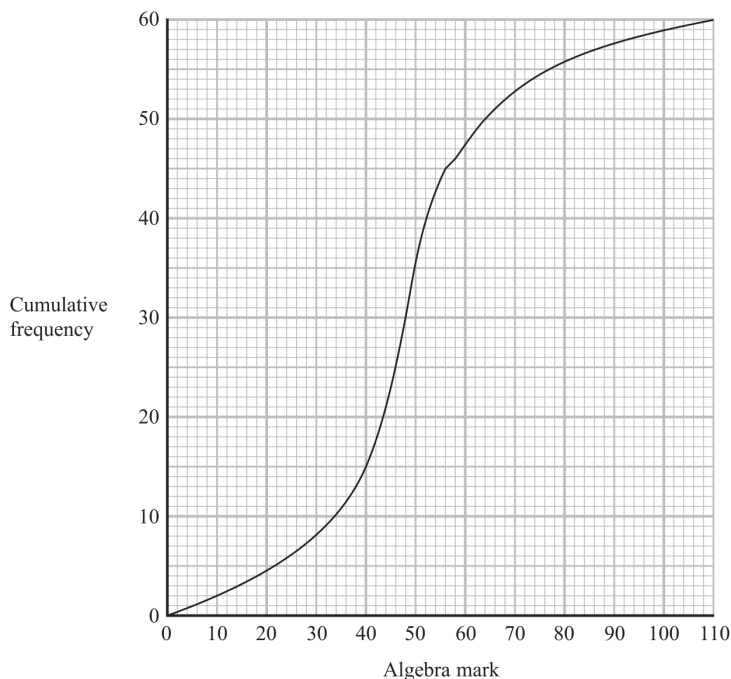
Q25**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q14

CH 6 • TOPIC 3 8 marks

- 14 A group of 60 students each sat an algebra test and a geometry test.
Each test was marked out of 110

The cumulative frequency graph gives information about the marks gained by the 60 students in the algebra test.



- (a) Use the graph to find an estimate for the median mark in the algebra test.
.....
- (b) Use the graph to find an estimate for the number of students who gained 58 marks or less in the algebra test.
.....
- (c) Use the graph to find an estimate for the interquartile range of the marks gained in the algebra test.
.....

The interquartile range of the marks gained in the geometry test is 9

Luis says

“The students’ marks are more spread out in the algebra test than in the geometry test.”

- (d) Is Luis correct?
Give a reason for your answer.
.....
.....

To be awarded a grade A in the algebra test, a student had to gain a mark greater than 64

Two students are to be selected at random from the 60 students in the group.

- (e) Use the graph to find an estimate for the probability that both of these students were awarded a grade A in the algebra test.
.....

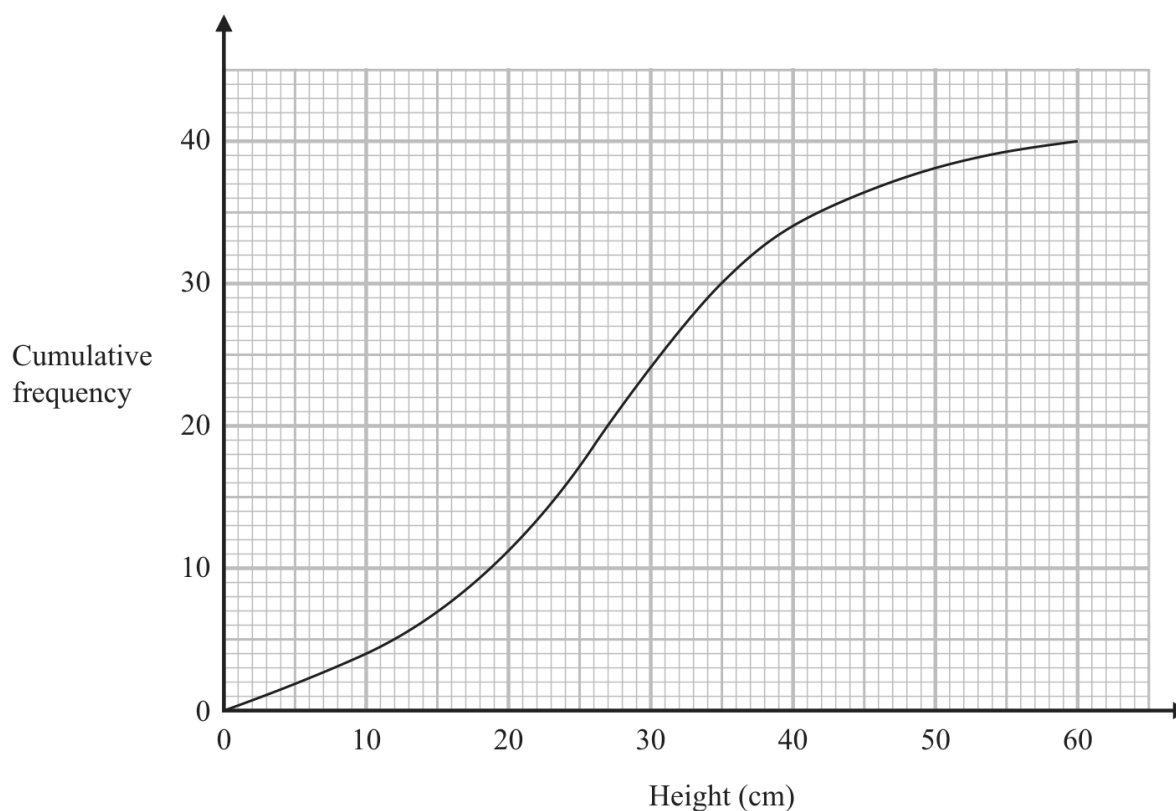
(Total for Question 14 is 8 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q14

CH 6 • TOPIC 3 5 marks

- 14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



- (a) Use the graph to find an estimate for the median height.

..... cm

- (b) Use the graph to find an estimate for the interquartile range of the heights.

..... cm

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

- (c) Work out the total amount of money Greta receives for the premium plants.

..... euros

(Total for Question 14 is 5 marks)

03 More Than, Less Than: Counts and Percentages

The graph always gives “less than or equal to”. For “more than x ”, read up at x and subtract from the total. Then turn it into a percentage or probability if asked.

WHEN YOU SEE → YOU DO

“more than 35 cm”

→ total – reading at 35

“between 42 and 52”

→ reading at 52 – reading at 42

“what percentage?”

→ $\frac{\text{count}}{\text{total}} \times 100$

“the top 20%” / “bags heavier than d ”

→ work backwards: count → cf → value

“two chosen at random”

→ $\frac{a}{n} \times \frac{a-1}{n-1}$

WORKED MODEL • NOVEMBER 2023 P1H

80 adults. Percentage who took longer than 50 seconds?

1 up at 50 → about 75

read

2 $80 - 75 = 5$

more than

ANSWER $\frac{5}{80} \times 100 \approx 6\%$

IDEA 03

Practise: More Than, Less Than: Counts and Percentages

My questions first, then selected practice, then real past papers.

Q27**DR ESLAM**

More than

CH 6 • TOPIC 3 2 marks

The graph (120 people) reads 87 at 45 minutes. How many took more than 45 minutes? What percentage?

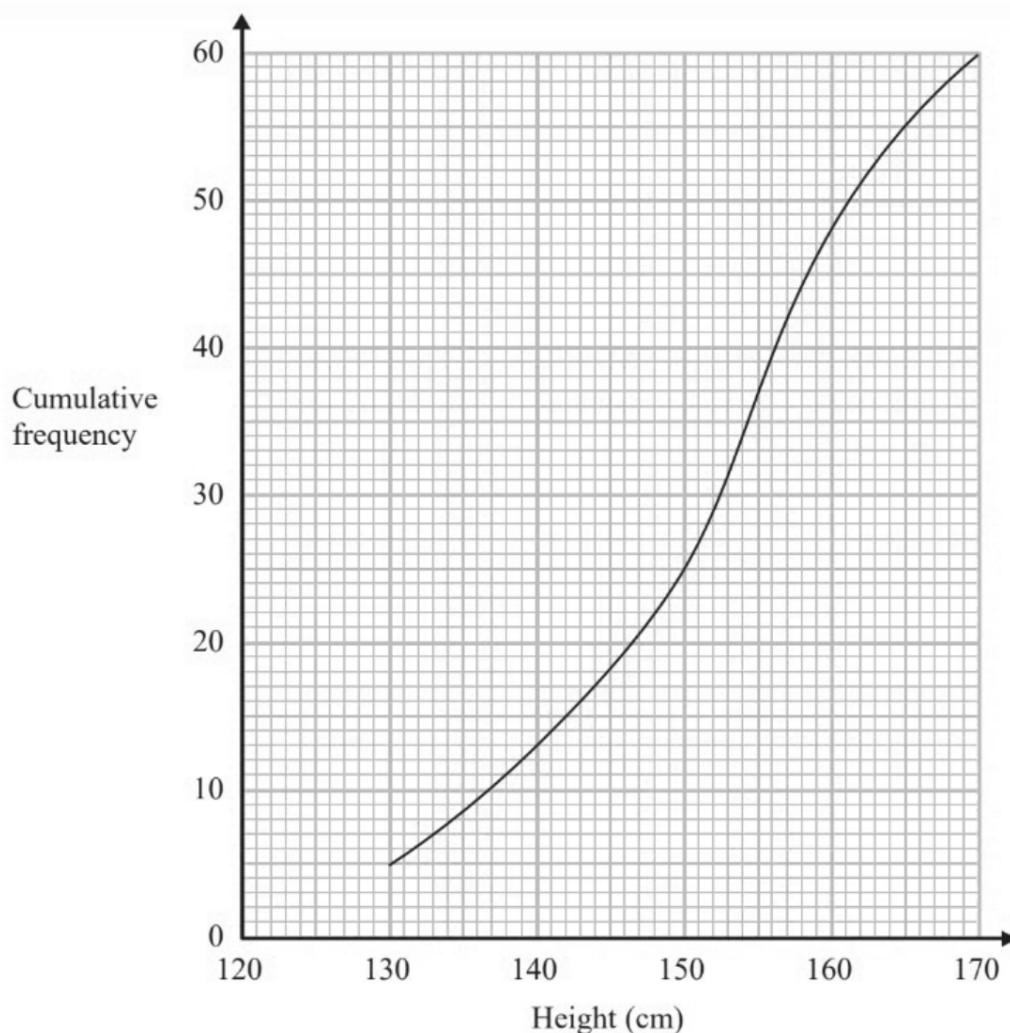
SELECTED PRACTICE

Q28**PRACTICE**

Practice • Easy

CH 6 • TOPIC 3 2 marks

5 The cumulative frequency graph shows some information about the heights, in cm, of 60 students.



Work out an estimate for the number of these students with a height greater than 160 cm.

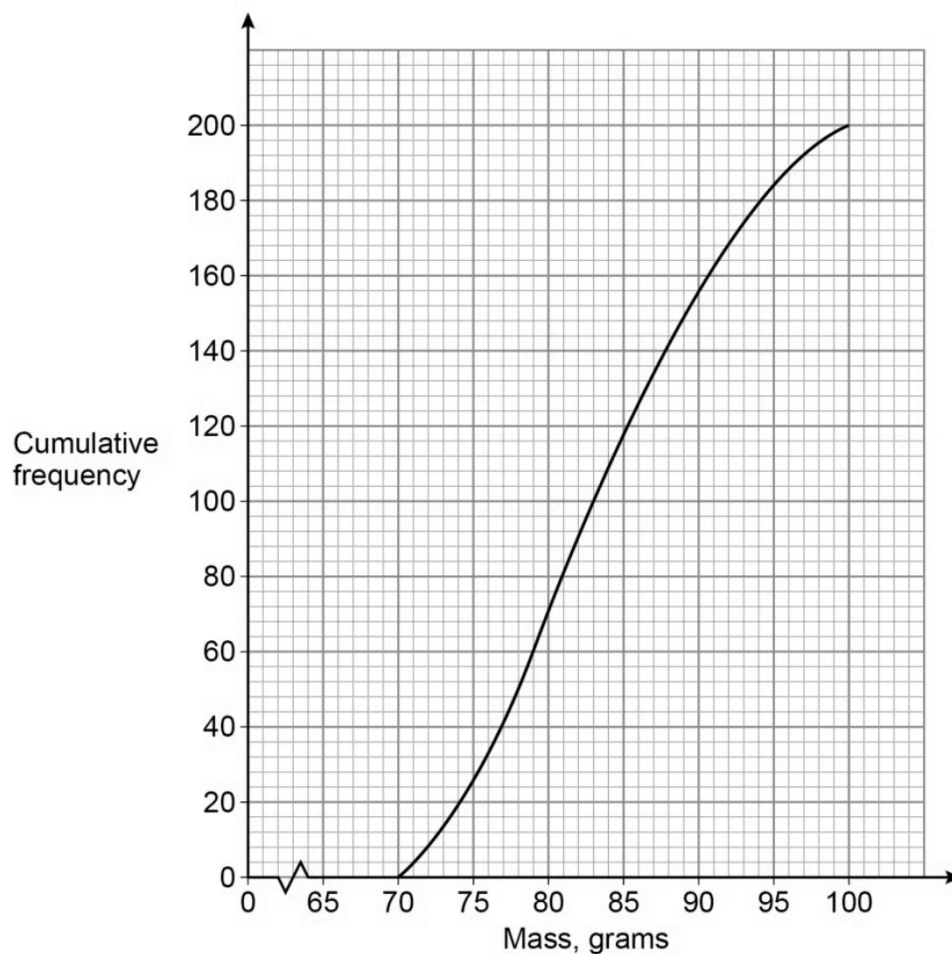
(2 marks)

Q29**PRACTICE**

Practice • Hard

CH 6 • TOPIC 3 4 marks

10 (a) The cumulative frequency graph shows information about the masses of 200 apples.



Estimate the median mass.

.....grams

(1 mark)

(b) Apples with mass 90 grams or less cost 32p each.

Apples with mass more than 90 grams cost 39p each. Estimate the **total** cost of the 200 apples.

£.....

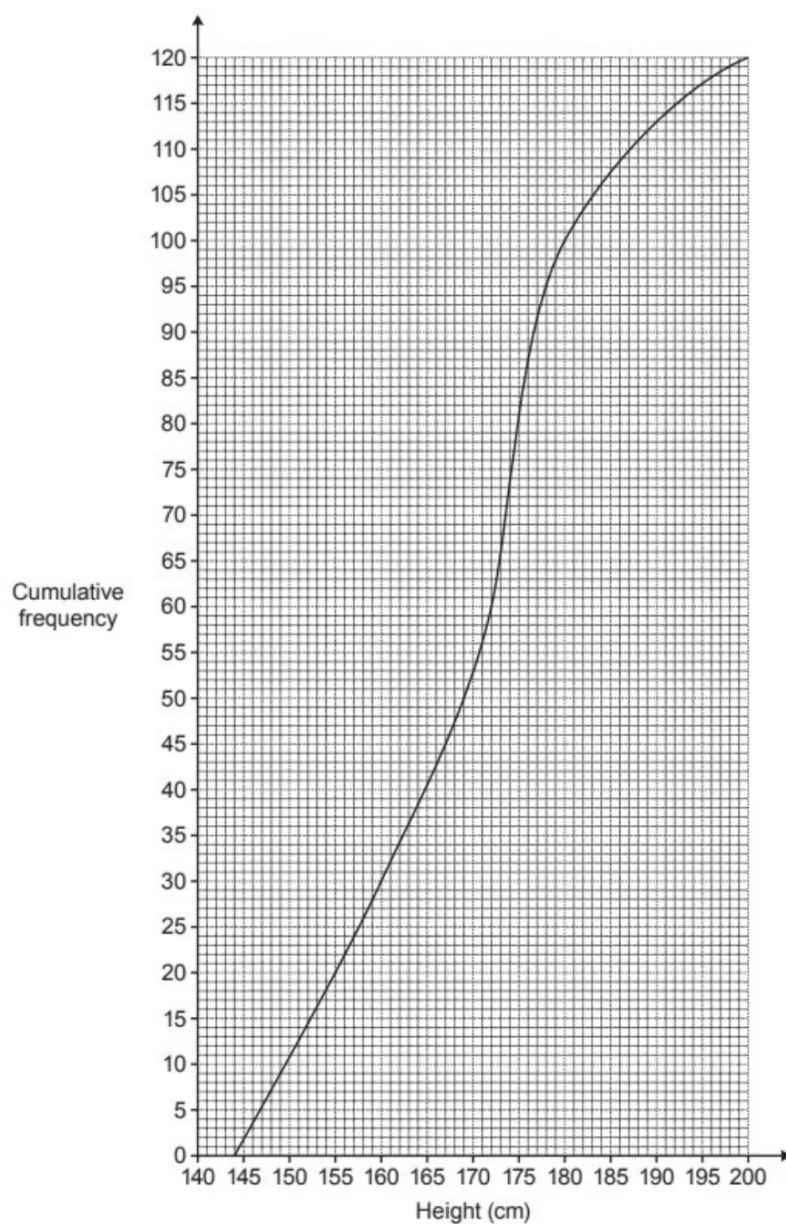
(3 marks)

Q30**PRACTICE**

Practice • Hard

CH 6 • TOPIC 3 **3 marks**

14 The cumulative frequency graph shows the distribution of the heights of members of a rowing club.



Calculate the percentage of the members who are at least 180 cm tall.

(3 marks)

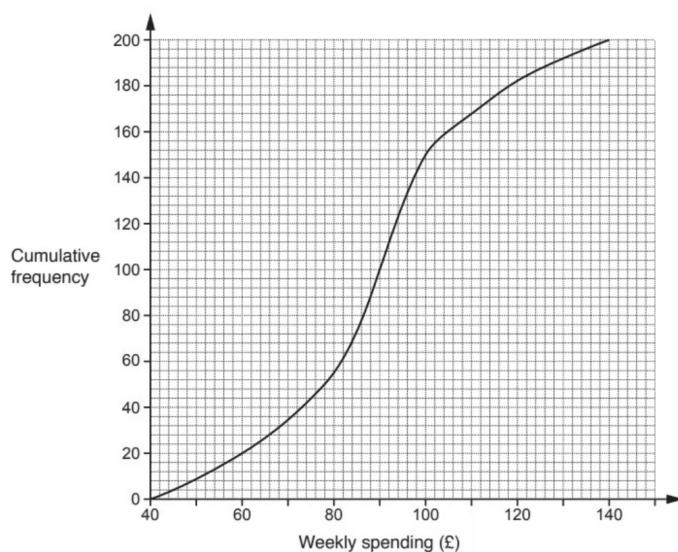
Q31**PRACTICE**

Practice • Hard

CH 6 • TOPIC 3 9 marks

- 16 (a)** Iqrah carries out a survey of 200 families in the **north** of England on their weekly spending on food.

The cumulative frequency diagram summarises the results.



Find

- i) the median,

£ [1]

- ii) the interquartile range.

£ [2]
(3 marks)

- (b)** Iqrah says

15% of these families spent over £120.

Is her statement correct?

State the evidence you have used in making your decision.

(2 marks)

- (c)** In a survey of 200 families in the **south** of England, the median weekly amount spent on food was £84 and the interquartile range was £28.

Make two comparisons between the weekly amounts spent on food in the north of England and the south of England. State the evidence you have used in making your comparisons.

1

2

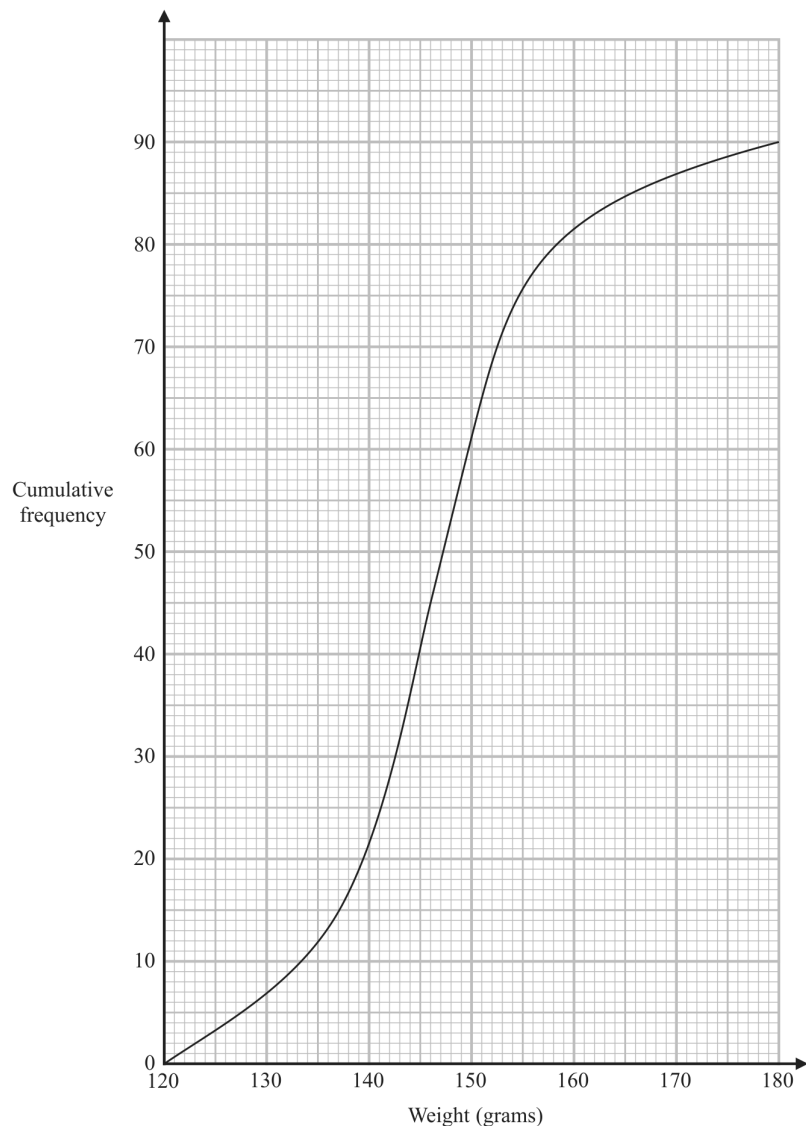
(4 marks)

PAST PAPERS

Q32 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 1H • Q11

CH 6 • TOPIC 3 5 marks

- 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



- (a) Find an estimate for the median of the weights of these bags of sweets.

..... grams

Roberto sells the bags of sweets to raise money for charity.

Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

- (b) Find the value of d .

$d =$

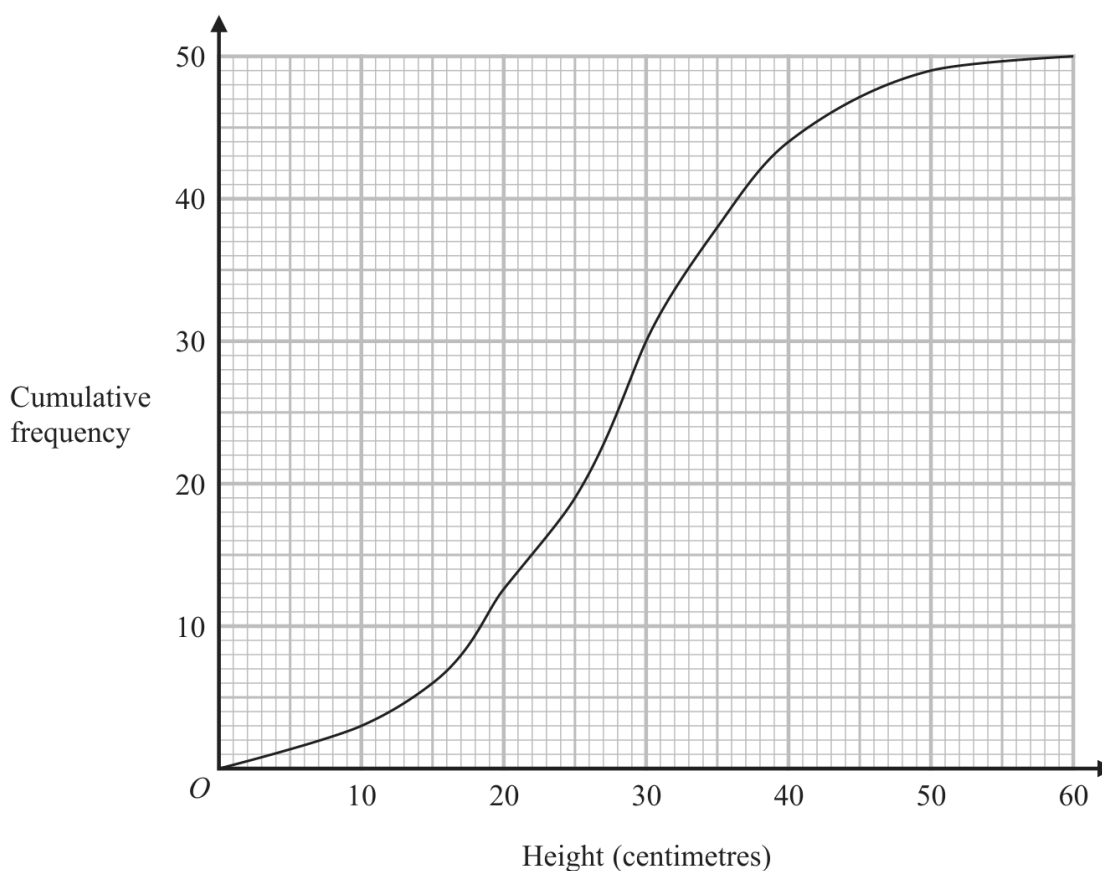
(Total for Question 11 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q12

CH 6 • TOPIC 3 4 marks

- 12 The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



- (a) Use the graph to find an estimate for the median height of these plants.

..... centimetres

- (b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leq 40$

.....

- (c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

.....

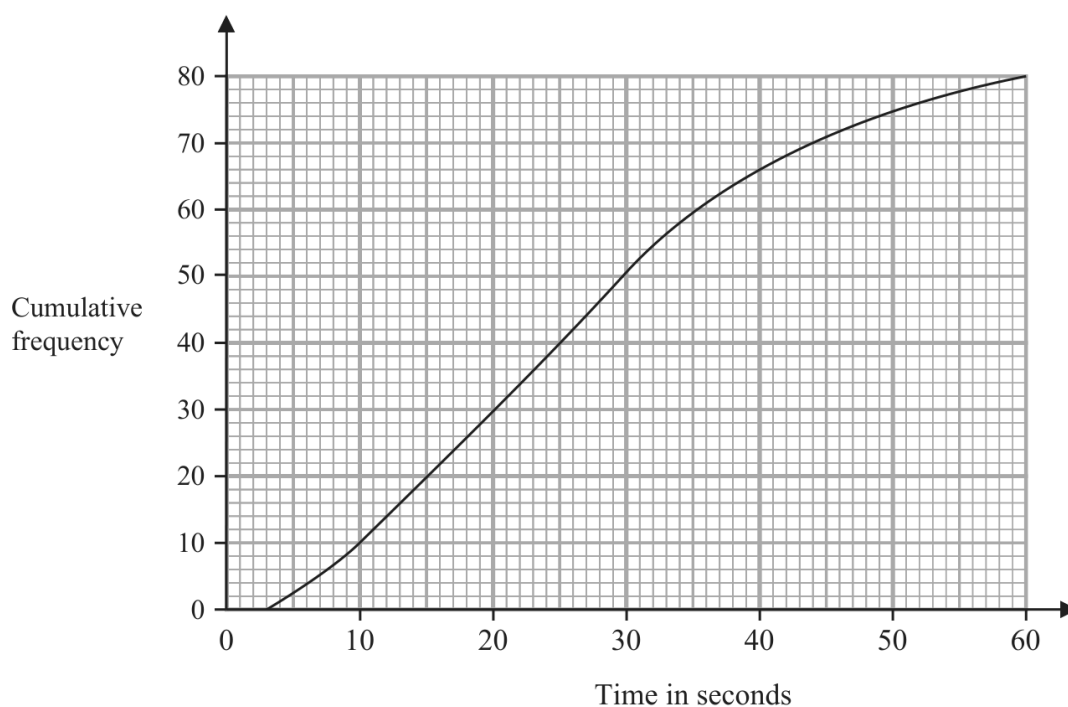
(Total for Question 12 is 4 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q13

CH 6 • TOPIC 3 4 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds

- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

..... %

(Total for Question 13 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|--|
| 01 | Graph | Running total; plot at upper bounds; smooth curve. |
| 02 | Median, IQR | Read at $\frac{n}{2}$; $\frac{n}{4}$ and $\frac{3n}{4}$; subtract. |
| 03 | More than | Total – reading; then % or probability. |

BOTTOM LINE Running total → upper bounds → across and down. More than = total – reading.

Answers

Cumulative Frequency · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CUMULATIVE FREQUENCY TABLES AND GRAPHS

- Q1** Dr Eslam • Running total 6, 20, 40, 47, 50; points (10, 6), (20, 20), (30, 40), (40, 47), (50, 50)
- Q2** Practice • Easy (a) 2, 11, 34, 65, 92, 100 (b) plot at the upper bounds (c) $100 - 80.5 \approx 20$ lorries
- Q3** Practice • Easy 2, 7, 11, 17, 20
- Q4** Practice • Medium (a) 4, 20, 56, 80, 94, 100 (c) median about 48 min (d) about $100 - 85 = 15$ people
- Q5** Practice • Hard (a) about £24 thousand (b) frequencies 14, 12, 10, 14, 10 (c) $1710 \div 60 = 28.5$ thousand (d) the mean uses midpoints of wide classes; the median is read directly
- Q6** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q11 Median about 37 years
- Q7** Edexcel 4MA1 • January 2021 • Paper 1HR • Q11 Cumulative frequencies 8, 23, 40, 68, 101, 120, median about 18.6 minutes, percentage about 27%
- Q8** Edexcel 4MA1 • May 2021 • Paper 2H • Q12 Median 11.6 km, IQR about 10.2 km
- Q9** Edexcel 4MA1 • November 2021 • Paper 1H • Q11 About 27 runners
- Q10** Edexcel 4MA1 • January 2022 • Paper 1H • Q12 Cumulative frequencies 7, 33, 57, 71, 78, 80, median about 23 minutes, probability about 0.095
- Q11** Edexcel 4MA1 • June 2022 • Paper 2HR • Q11 Median about 27.4 years, IQR about 25.1 years, about 11 people older than 55
- Q12** Edexcel 4MA1 • June 2022 • Paper 2H • Q13 Cumulative frequencies 7, 32, 52, 66, 74, 80, median about 34 years, about 42 people going to work
- Q13** Edexcel 4MA1 • May 2023 • Paper 1HR • Q14 Median about 74.3 min, IQR about 22.6 min
- Q14** Edexcel 4MA1 • June 2024 • Paper 2HR • Q11 IQR about 12.3, $W \approx 17.7$, and $\frac{11}{60}$
- Q15** Edexcel 4MA1 • June 2024 • Paper 2H • Q13 IQR about 21 km, and about 8 teachers

IDEA 02 • MEDIAN, QUANTILES AND IQR FROM THE GRAPH

- Q19** Dr Eslam • Positions 40; 20; 60
- Q20** Practice • Easy (a) about 128 g (b) $133 - 122 \approx 11$ g
- Q21** Practice • Medium (a) about 57 g (b) No: the exact smallest and largest weights are unknown (c) $60 - 49 = 11$, $\frac{11}{60} \approx 18\% < 25\%$
- Q22** Practice • Hard (a) $123 - 102.5 \approx 20.5$ min (b) $80 - 56 = 24$ films = 30%: Clare is not correct
- Q23** Edexcel 4MA1 • January 2020 • Paper 1HR • Q12 Median about 22.5 minutes, IQR about 12 minutes. Hospital A is quicker and more consistent
- Q24** Edexcel 4MA1 • January 2020 • Paper 1H • Q12 IQR about 18 minutes, and about 70 men
- Q25** Edexcel 4MA1 • January 2022 • Paper 2HR • Q14 Median about 49, about 45 students, IQR about 17, Luis is correct, probability about 0.025
- Q26** Edexcel 4MA1 • November 2025 • Paper 2H • Q14 Median about 26 cm, IQR about 16 cm, and 180 euros

IDEA 03 • MORE THAN, LESS THAN: COUNTS AND PERCENTAGES

- Q27** Dr Eslam • More than
 $120 - 87 = 33$; 27.5%
- Q28** Practice • Easy
About 12 students
- Q29** Practice • Hard
(a) about 83 g (b) about £67.08
- Q30** Practice • Hard
 $120 - 100 = 20$ of 120: 16.7%
- Q31** Practice • Hard
(a) median about £90, IQR $100 - 78 = £22$ (b)
Not correct: 15% spend over about £112, not £120
Median about 146 g, and $d \approx 150$
- Q32** Edexcel 4MA1 • January 2021 • Paper 1H • Q11
Median about 27 cm, about 13 plants, about 13 plants
- Q33** Edexcel 4MA1 • May 2023 • Paper 1H • Q12
Median about 25 seconds, and about 6%
- Q34** Edexcel 4MA1 • November 2023 • Paper 1H • Q13